FREDHOLM REDUCTION: 12 MATHEMATICAL STEPS

Exact transport → compact model → WH/Mellin classes → local finite symbol

READ LEFT TO RIGHT, TOP TO BOTTOM

01 Mixed star Γ

$$\gamma_k(x)=(x-ic_k)^{-1}\ (k\le s),\quad \gamma_k(x)=e^{i\beta_k}/x\ (k>s)$$

02 Real algebra B

$$\mathfrak{B}=\mathrm{alg}\{aI,S_\Gamma,C\}$$

03 N = A + BC

$$N=A+BC,\quad A,B\in\mathfrak{A},\quad Cf=\bar{f}$$

04 Exact 2×2 lift

$$\widetilde{N}=\Theta(A+BC)$$

05 Isometry σ

$$\sigma: L^p(\Gamma,\widetilde{w})\to L^p_m(\mathbb{R}_+,w_\delta)$$

06 Exact matrix T

$$\sigma S_\Gamma\sigma^{-1}=[T_{k,j}]_{k,j=1}^m$$

07 Model mod K

$$\sigma S_\Gamma\sigma^{-1}=[S_{k,j}]+K,\quad K\in\mathcal{K}$$

08 WH + Mellin blocks

$$S_{k,j}\in\{\chi_\infty W(b)\chi_\infty, M^0(c), \chi_\infty M^0(c)\chi_\infty\}$$

09 Local domain Ω-hat

$$\omega=(\xi,\eta,x)\in\widehat{\Omega}$$

10 Finite m×m symbol

$$\Phi(A)(\omega)=[A](\omega)\in M_m(\mathbb{C}),\quad \omega\in\widehat{\Omega}$$

11 Finite 2m×2m symbol

$$\mathcal{Y}(N)(\omega)\in M_{2m}(\mathbb{C})$$

12 Fredholm determinant

$$N\ \text{Fredholm}\Leftrightarrow \det\mathcal{Y}(N)(\omega)\neq 0\ (\forall\omega\in\widehat{\Omega})$$

Karlovich 2025: (1.1), (3.5)–(3.28), (6.5), Theorems 6.2 and 7.2. The final determinant belongs to the 2m×2m symbol on Ω-hat.